

RAHUL KARTHIK

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PROFESSIONAL SUMMARY

Data-focused professional with a background in physics and analytics, skilled in Python scripting and SQL. Experienced in building workflows, data models, visual dashboards, and internal analytics tools. Strong communicator adept at translating technical data for operational use. CAPM and CPMAI credentials from PMI. US Citizen.

EDUCATION

Stony Brook University - SUNY

Master's, Physics

August 2023 - May 2025

GPA: 3.6

- Relevant coursework: Methods of Mathematical Physics

Stony Brook University - SUNY

Certification, Data Science

August 2023 - May 2025

GPA: 3.6

- Relevant coursework: Intro. to Computational Science, Probability and Statistics, Foundations of Quantitative Finance, Foundations of Science Communication

Shiv Nadar Institute of Eminence (SNIOE)

Bachelor's, Physics

July 2018 - May 2022

GPA: 3.4

- Relevant coursework: Introduction to Computational Physics I & II, Machine Learning through R, Big Data Modelling and Integration

CERTIFICATIONS

Certified Associate in Project Management | Project Management Institute (February 2021)

Cognilytica CPMAI+E v7 | Project Management Institute (February 2025)

The Data Science Course: Complete Data Science Bootcamp 2024 | 365 Careers, Udemy (December 2024)

Exploratory Data Analysis in SQL | DataCamp February 2025

Power BI Essential Training (NASBA certified) | Gini von Courter, LinkedIn Learning (March 2025)

SKILLS

Technical Skills: Python (NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, PyTorch), Machine Learning Models (Regression, Decision Trees, XGBoost, Neural Nets), SQL, Git, R, Power BI, Tableau, Excel/Numbers/Sheets, Scala, Snowflake, Apache Spark, Data Analytics, Data Mining

Soft Skills: Technical Writing, Critical Thinking, Problem Solving, Communication, Cross-functional Collaboration, Adaptability, Agile Familiar

Tools: VS Code, PyCharm, GitHub, MS Office Suite (Excel, Word, Powerpoint, Sharepoint)

Languages: Hindi, Tamil

PROFESSIONAL EXPERIENCE

Stony Brook University - SUNY

Graduate Researcher

Stony Brook, NY, USA

August 2023 - Present

- Developed Python-based ETL workflows to process 2TB+ datasets; aligned with industrial-scale database pipelines.
- Created modular, version-controlled data tools using Git and Jupyter for reproducible research.
- Supported a publication involving real-time data visualization and complex metric tracking.
- Collaborated with multi-disciplinary teams — mirroring internal stakeholder collaboration for operational insights.

SolutionsMax Technology Services

Social Media and Marketing Analyst

Visakhapatnam, AP, India

January 2023 - July 2023

- Created analytics-driven video content strategy with market research KPIs that improved engagement by 40%.
- Reinforced storytelling with data, including trends, metrics, and visualizations.

SolutionsMax Technology Services

Visakhapatnam, AP, India

R&D Intern

July 2022 - January 2023

- Developed and proposed production and packing-line solutions, reducing time at pharmaceutical traceability compliance checkpoints in Indian factories, leading to a 3x speedup in operations.
- Modeled system layouts in AutoCAD and optimized production flow using statistical decision-making frameworks.

Shiv Nadar Institute of Eminence (SNIOE)

Delhi, India

Undergraduate Researcher

September 2021 - May 2022

- Conducted advanced data modeling for quantum systems; automated reporting using SQL and Python
- Reduced manual reporting time by 30% through workflow automation.
- Presented research findings at IIT Madras Journal Club using data dashboards and probabilistic models.

AhaGuru Education Technology Pvt. Ltd.

Chennai, TN, India

Intern with Education Research and Content Creation Teams

June 2020 - September 2020

- Researched optimal teaching and presentation techniques to improve conceptual understanding, enhancing content quality for competitive entrance exams.
- Developed graded materials and contributed to the Content Creation Team, refining preparation materials.

AhaGuru Education Technology Pvt. Ltd.

Chennai, TN, India

Intern with Content Creation Team

May 2019 - August 2019

- Compiled comprehensive revision booklets on topics for IIT JEE and NEET exams, providing students with essential review.
- Designed and conducted 15 interactive problem-solving sessions for grades 10 and 11, enhancing students' understanding and confidence in core mathematics topics.

International Book: "Say Yes to Project Success"

Chennai, TN, India

Research and Content Review

May 2017 - August 2017

- Researched global project success principles using a unique "poll of polls" approach, enriching the book's evidence base.
- Analyzed 42 independent Project Management surveys and studies to identify key project failure factors.

PROJECTS & OUTSIDE EXPERIENCE

Publication: "The Roman View of Strong Lenses"

Stony Brook, NY, USA

Co-author

September 2023 - April 2025

- Co-author on a 20-member collaboration presenting the first yield simulation of 160,000 strong-lens systems in Roman's High Latitude Wide Area Survey.
- Integrated COSMOS F814W images into a Bayesian neural-net pipeline for dark-matter inference, handling data ingestion, cleaning, and feature selection.
- Matched SkyPy source properties to real-image cutouts via nearest-neighbor algorithms to drive large-scale training.
- Extended SLSim: added real-image interpolation, refined PSF convolution, and implemented noise-reduction routines for improved simulation fidelity.

Event-Driven Earnings Impact Predictor

New York, NY, USA

Lead (Passion project)

May 2025 - June 2025

- Assembled multi-ticker S&P 500 earnings dataset (2004–2024) via Wikipedia scrape & yfinance (300 events)
- Engineered 15 event-based features: multi-horizon returns, moving averages, volatility (std/ATR), RSI, volume spikes, momentum, calendar flags & surprise%
- Built time-series RF, GBM & MLP models achieving 60% accuracy (AUC 0.60); surprise% & volatility ranked highest in importance
- [Link to project](#)

Indian Premier League (IPL) Scorecard Predictor

Stony Brook, NY, USA

Lead

April 2024 - May 2024

- Used Python and Bayesian analysis to model team cricket performance using 400+ event datasets.
- Demonstrated cross-domain potential for analytics and forecasting models.
- [Link to project](#)

Heart Disease Analysis and Prediction

Delhi, India

Co-Lead

February 2021 - April 2021

- Built ensemble ML models achieving 92% accuracy; applied to real-world healthcare indicators.
- Demonstrated feature optimization and early-stage anomaly detection potential
- [Link to project](#)